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Classification of Papaya Fruit Quality Based on Shape, Texture, and Sugar Level Using Artificial Neural Network Algorithm

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Abstract. Papaya, a tropical plant commonly found in regions such as the South Pacific and other tropical areas, thrives in low-temperature environments, making it ideally suited for tropical climates. In addition to its popularity as a fresh fruit, papaya is extensively used in various preparations. However, one of crucial post-harvest processes to sort its quality is still conventionally carried out by manual human labor, presenting inherent limitations and shortcomings. This study employs the artificial neural network method to classify the quality of papaya fruit based on three features which are shape, texture and sugar content levels to address this issue. The research encompasses distinct stages, including image acquisition, preprocessing, segmentation, morphological operations, feature extraction, and classification. The dataset used comprises 600 data points, divided into 480 for training and 120 for testing. The outcomes reveal that employing the artificial neural network algorithm based on these features can achieve a commendable accuracy of 90%, with an ME (Mean Error) of 10. The results indicating that by using these features, the papaya quality classification system accomplished within a remarkably swift time of only 1032.705593 seconds.

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INTRODUCTION

Papaya, a tropical plant, is abundant in the South Pacific and other tropical regions and thrives in warmer climates. It is well-suited for cultivation in tropical environments due to its preference for lower temperatures [1]. In Indonesia, various papaya types are commonly grown, including watermelon paris papaya, dampit papaya, Cibinong papaya, orange papaya, mini papaya (Hawaii), solo papaya, sun rise papaya, and California papaya. Additionally, the fruit research center has introduced several new papaya varieties, such as papaya carindo, sari gading, and sari raon. Papaya is not only enjoyed as a fresh fruit but also finds extensive application in various food products. The food industry often incorporates papaya into items such as jams, syrup drinks, pharmaceuticals, and cosmetics [2]. However, papaya's perishable nature presents a challenge when marketing it as a fresh fruit that needs to remain in optimal condition for consumers.

The fruit is spherical, yellow, and can reach up to 10 meters in height, with a tree trunk that is hollow. The color of the fruit transitions from light green to brownish as it ripens [3]. The skin of the papaya fruit changes from green to yellow, and the flesh changes from white to red as it ripens [4]. This fruit contains two biologically active compounds, along with numerous beneficial nutrients, particularly provitamin A. The vitamins within this fruit encompass organic compounds essential for metabolic reactions in cell growth and overall health. In a 100-gram serving of papaya fruit, you can find 12.4 grams of carbohydrates, 23 mg of calcium, 12 mg of phosphorus, 1.7 mg of iron, 110 mcg of retinol, 0.04 mg of thiamine, and 78 mg of vitamin C [5].

Papaya leaves are rich in chemical compounds such as flavonoids, dextrose, levulose, papain enzymes, saccharose, and more, which possess antibacterial properties. Consequently, many people utilize papaya leaves as traditional remedies, including treatments for malaria, appetite enhancement, deworming, kidney stone relief, menstrual bleeding, and pain alleviation [6]. The popularity of papaya leaf treatments lies in their availability regardless of weather or season [7]. In addition to accessibility, using natural plant-based products for treatment offers significant advantages, including lower costs and relatively fewer side effects compared to synthetic drugs with potential long and short-term adverse effects [8] [9][10]. Data from the World Health Organization (WHO) suggests that approximately 80% of

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people worldwide still prefer plant-based ingredients in medicinal preparations [11]. Hence, supporting technological advancements in agriculture is essential to produce higher-quality crops [12].

As per information from the Food and Agriculture Organization (FAO), global papaya fruit production reaches 12.5 million tons annually, with Indonesia ranking as the fifth largest papaya fruit producer in the world, following Brazil, Nigeria, India, and Mexico [13]. Therefore, the demand for papaya fruit for export necessitates effective quality management to compete with papaya producers from other countries involved in papaya fruit exports [14].

However, in the process of assessing the quality of papaya fruit, a crucial post-harvest concern, individuals still conventionally and manually perform the quality assessment. This approach indeed comes with limitations and shortcomings. Manual sorting demands significant labor, and the perception of quality can vary among individuals which results in inconsistency of the quality assessment process.

Previous research, such as that conducted by Feri Wibowo and Agus Harjoko in 2017, focused on papaya fruit quality classification based on GLCM texture characteristics using artificial neural networks and achieved an accuracy rate of 86.11% [15]. Likewise, research by A. Isatul Masruroh, Sorikhi, and Achmad Syauki in 2023, which explored the maturity level of California papaya in the HSV color space using the K-Nearest Neighbors method, yielded an accuracy rate of 86.66% [16]. The previous research becomes evident that it relied on a single feature for quality assessment, suggesting that there is room for improvement in terms of accuracy. One approach to enhance accuracy is to increase the volume of data used for training and testing.

Therefore, this study proposes the classification of papaya fruit quality based on both texture and sugar content using artificial neural networks, incorporating additional features to aid in quality assessment. The study classifies papaya into three categories: good, medium, and bad. With this approach, the designed system aims to achieve a higher level of accuracy.

The method proposed in this research employs artificial neural networks, which are mathematical models inspired by the structure and function of human biological neural networks. For instance, the elements in artificial neural networks can be connected in parallel, enabling cooperative operation. Artificial neural networks are often likened to artificial brains, as they have the capacity to draw conclusions based on the information they receive [17]. The utilization of the Artificial Neural Network method for classifying papaya fruit quality is expected to enhance the accuracy of the assessment process, resulting in more precise and effective accuracy rates within the good, medium, and bad categories.

METHODS

The method used in this research was artificial neural network. In using the model, several stages were needed, namely, image acquisition stage, preprocessing stage, segmentation stage, morphology, feature extraction, and classification. can be seen in Fig. 1.

- Acquisition; the data utilized in this research comprised 600 image datasets, which is twice the number used in previous studies. The data source consisted of primary data collected directly by the researchers. The data was divided into training and test sets, with 80% allocated for training (480 data points) and 20% for testing (120 data points).
- Preprocessing; preprocessing is a stage of image improvement to remove noise (parts that are not needed) in order to increase the probability of success in image processing[18]. At this stage, preprocessing was carried out on the papaya fruit dataset by increasing the brightness of the original image, increasing the contrast in the original image, and reducing the image from RGB.
- Segmentation; following the completion of the image preprocessing stage, the process advanced to segmentation, where the objective was to distinguish the area between the object and the background. This separation facilitated a more straightforward analysis[19]. In this study, the Otsu segmentation method was employed, which involves determining the threshold value through histogram-based differentiation between objects and backgrounds [20].
- Morphology; morphological operations in digital image processing are the process of manipulating an object in the original image if the results of the segmentation stage are not perfect. After the morphological operation phase, the procedure advanced with dilation and erosion processes, followed by closing operations, hole filling, and the removal of noise from image objects. These actions were undertaken to attain a precise and distinct segmentation result.

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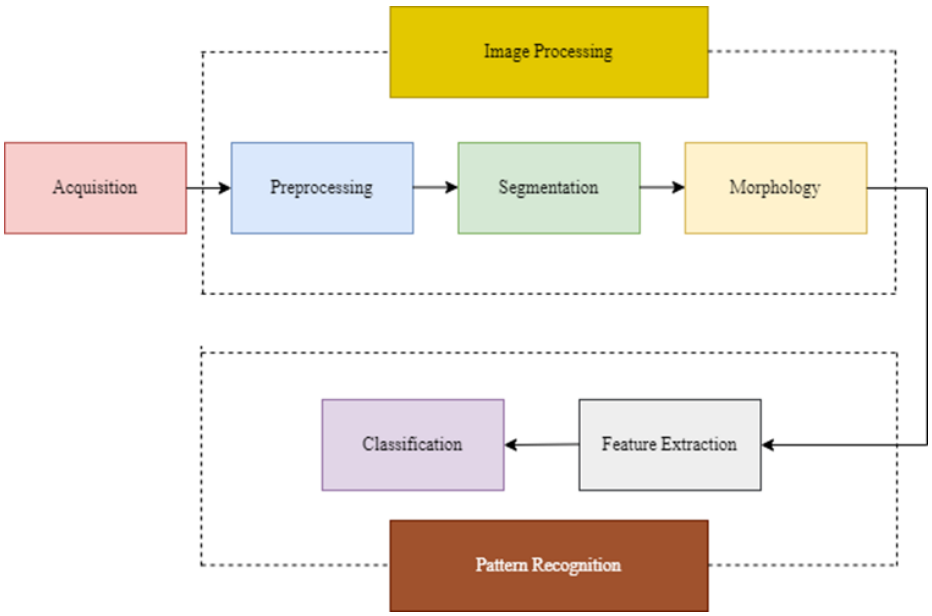

FIGURE 1. Method stages.

- Feature extraction; after the morphological operations, the object in the image became clear and distinct. Subsequently, feature extraction was conducted in the object area. In this phase, the GLCM method was employed for feature extraction, pinpointing texture shape patterns within the image. This process generates output through the extraction of color features, specifically using the mean value. These extracted features will act as a standard for quality classification, considering both texture and sugar content levels.
- Classification; during this phase, the classification procedure was executed to sort and group every pixel in the digital image into specific classes or categories, depending on the attributes or features extracted from the image. The primary objective of classification was to offer a more meaningful interpretation of the image, enabling us to recognize and comprehend the objects or features depicted in it.

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RESULTS AND DISCUSSION

This study uses a dataset of 600 images, which are categorized into three classes: good, medium, and bad, as an example is shown by Fig. 2. Good quality has a clean skin surface with a smooth texture and a dominant orange/yellow color. Medium quality has clean skin but there are slight wounds in the texture and the dominant color is yellow and a little green. Poor quality has moldy skin and a texture that is filled with many wounds and is dominated by a brownish yellow color or a more dominating green color.

(a) (b) (c)

FIGURE 2. Datasets, (a) Good, (b) Medium, and (c) Bad

After performing image acquisition by clustering the dataset, the preprocessing stage is next. At this stage, channel conversion is carried out on the RGB image to determine the appropriate channel used in the segmentation stage. However, before the segmentation stage is carried out, image enhancement is first carried out by increasing the brightness and increasing the contrast in the image. The result is shown Fig. 3.

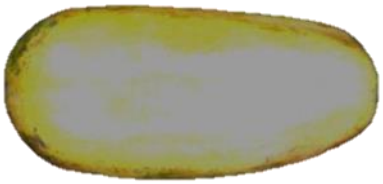

FIGURE 3. Preprocessing result.

Next, the segmentation stage is carried out, namely the separation of objects and backgrounds using the Otsu segmentation method. This method determines the threshold value by distinguishing objects and backgrounds based on histograms. if it look at the segmentation results in Fig. 4 there are still background areas that are detected as objects. Poor segmentation results will affect the feature extraction process, therefore refining the segmentation results at the morphological operation stage is carried out.

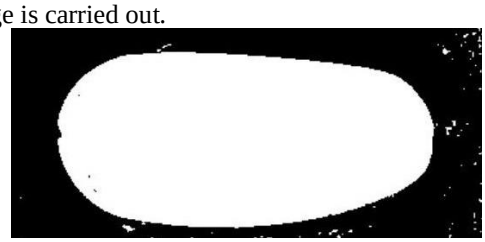

FIGURE 4. Segmentation result.

After the segmentation stage, morphological operations are performed. This involves creating disk strels of two sizes: strel 1 of 15 pixels and strel 2 of 30 pixels. Widening and closing operations are performed using the constructed strels, along with hole filling to close the hole objects. In addition, noise or unnecessary objects are removed to enhance the segmentation results, as shown in Fig. 5.

FIGURE 5. Morphology Results.

Next, the feature extraction stage is carried out. This research uses 2 training scenarios to determine the best feature extraction for the image. The first scenario is done by combining 2 features, namely RGB; Shape, RGB; Texture, HSV; Shape, HSV; Texture, LAB; Shape, and LAB; Texture, presented in Table 1.

TABLE 1. Training Scenario 1

No	Fitur	Accuracy (%)	ME	Running Time (sec)
1	RGB; Shape	62,50	37,5	2360.983060
2	RGB; Texture	62,50	37,5	2585.042020
3	LAB; Shape	79,58	20,42	3614.401513
4	LAB; Texture	74.16	25,84	4844.873443
5	HSV; Shape	55.20	44,8	316838
6	HSV; Texture	75.20	24.8	3665.457588

As depicted in Table 1 above, the LAB shape feature showed the highest accuracy rate of 79.58%, with an ME of 20.42% and a processing time of 3614.401513 seconds. After the first training scenario was completed, the second training scenario was implemented. This scenario involves combining 3 features namely RGB, Shape, and Texture features, as well as HSV, Shape, Texture, and LAB, Shape, Texture. The results of the training accuracy of scenario 2 are presented in Table 2.

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TABLE 2. Training scenario 2.

No	Fitur	Accuracy (%)	ME	Running Time (sec)
1	RGB; Texture; Shape	86.45	13,55	1667.697355
2	HSV; Texture; Shape	56.04	43,96	2733.732162
3	LAB; Texture; Shape	94,58	5,42	4148.897165

In Table 2, the highest accuracy is achieved by the LAB feature combined with Texture and Shape, showing an accuracy rate of 94.58% with an ME of 5.42% and a runtime of 4148.897165 seconds. While the combination of HSV, Texture, and Shape produces an accuracy of 56.04% with a ME of 43.96% and a runtime of 2733.732162 seconds. In addition, the RGB feature when combined with Texture and Shape produces an accuracy rate of 86.45% with a ME of 13.55% and a runtime of 1667.697355 seconds.

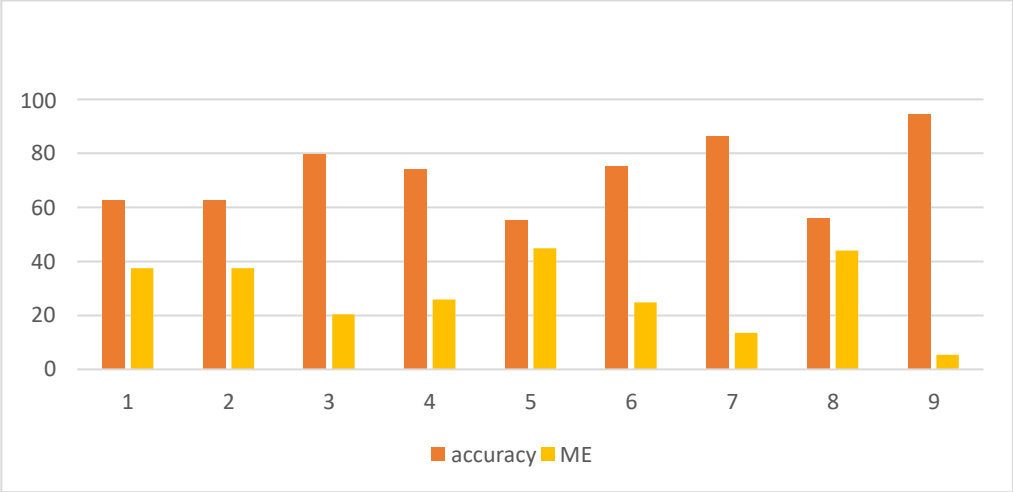

FIGURE 6. Comparison of index x feature combinations.

Figure 6 shows a comparison of the accuracy and misclassification error (ME) of all trials of feature combinations in training scenarios 1 and 2, it can be seen that the most effective combination is at index 9, namely the combination of LAB, Texture, and Shape features with an accuracy rate of 94.58% with a misclassification of 5.42. and index 7, namely the combination of RGB, Texture, and Shape features with an accuracy rate of 86.45% with a misclassification of 13.55. These two feature combinations are the most appropriate combinations for classifying papaya fruit quality and will be used in the testing phase.

TABLE 3. Testing.

No	Fitur	Accuracy (%)	ME	Running Time (sec)
1	LAB; Texture; Shape	88,33	11,67	1061.773806
2	RGB; Texture; Shape	90	10	1032.705593

Based on table 3, the testing stage is carried out with the results of a combination of 3 features using test data as much as 120 image data then divided into 40 data for good quality, 40 data for medium quality, and 40 for poor quality. By analyzing the test results above, it can be seen that the best accuracy level is found in the combination of RGB; Texture; Shape features with an accuracy level of 90% and an estimated time of 1032.705593 seconds.

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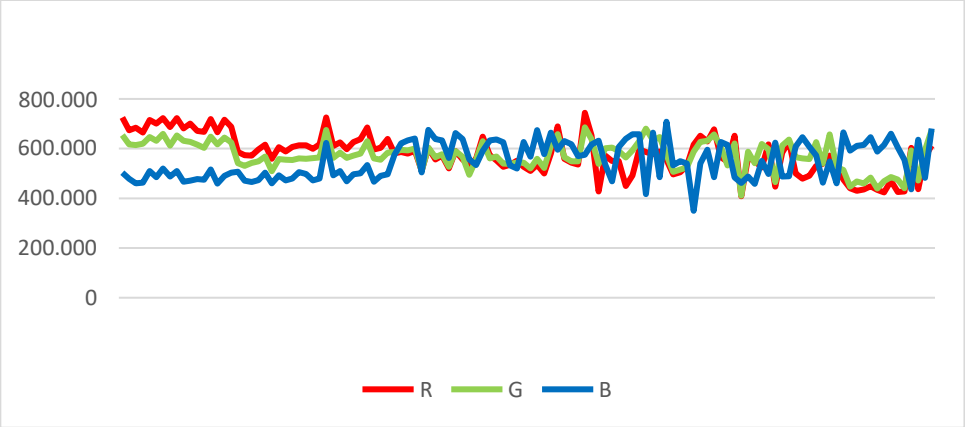

FIGURE 7. Comparison of index x feature combinations.

TABLE 4. Classification results.

Sugar Level	Texture	Shape	Test Image	Accuracy (%)
High	Good	Good	40	90%
Medium	Medium	Medium	40	90%
Low	Bad	Bad	40	90%

In Table 4 there are classification results from the testing stage. Based on this data, the classification of papaya fruit with good, medium, and poor quality using 40 test data each with an accuracy rate of 90%. Therefore, it can be concluded that the use of artificial neural network (JST) algorithms using digital image processing techniques for the classification of papaya fruit quality based on shape, texture, and sugar content levels can be done with an accuracy level that is considered very accurate.

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CONCLUSIONS

The application of artificial neural network methods in classifying papaya fruit quality according to its shape, texture, and sugar content level has yielded satisfactory results. The process involves stages such as image acquisition or dataset collection, preprocessing, segmentation, morphological operations, feature extraction, and classification. This study achieved a remarkable accuracy rate of 90% by utilizing a combination of RGB color, shape, and texture features, utilizing a total of 600 datasets. These datasets were categorized into three groups: good (174 datasets), medium (223 datasets), and bad (203 datasets). This Papaya Fruit Quality Classification System, employing the artificial neural network (JST) algorithm, is expected to streamline the papaya fruit sorting process for export activities. Moreover, this research is anticipated to make valuable contributions to the fields of Informatics Engineering and Agriculture, enhancing knowledge and understanding in these domains.

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